

# PHOENIX FITNESS CENTRE

Modern Gym & Personal Fitness Academy — Production System

Specification

System Status: **OPERATIONAL** (200 OK)  
Live Deployment:  
frontend-jet-psi-40.vercel.app  
Gym Email: phoenixgym.vkp@gmail.com  
Desk & Support: +91 8015552425

## 1. Executive Summary & Operational Status

This document provides the definitive architectural specification, operational walkthrough, and visual flowcharts for the **Phoenix Fitness Centre** web platform. The system operates as a unified management platform designed for gym owners and desk operators to handle member enrollments, subscription lifecycle management, automated progressive expiry warnings, 1-click renewals, and real-time revenue telemetry.

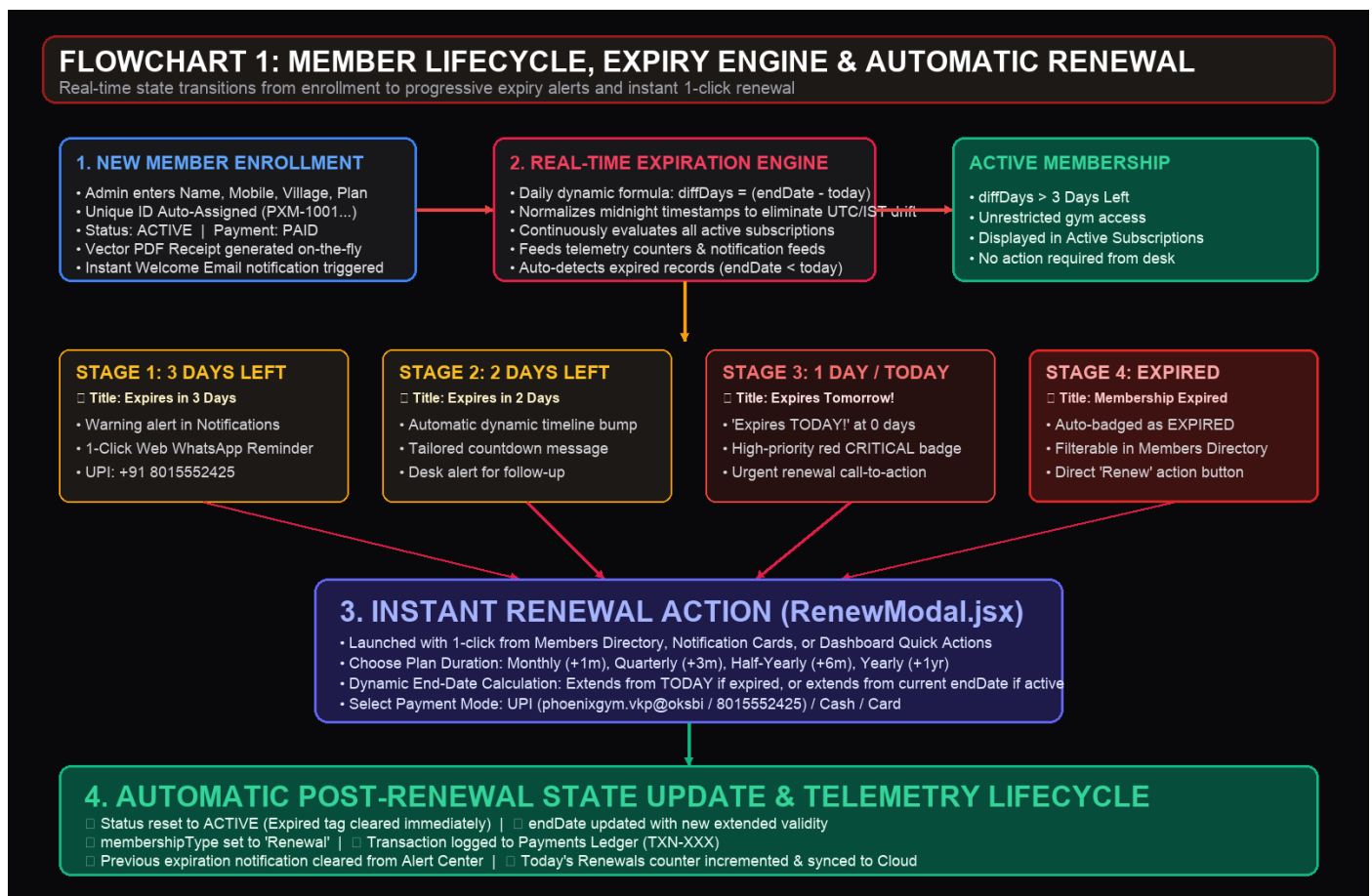
| Component / Area      | Operational State        | Target Endpoint / Details                   |
|-----------------------|--------------------------|---------------------------------------------|
| Vercel Web App        | Active / Production      | https://frontend-jet-psi-40.vercel.app      |
| Official Contact Line | Configured & Active      | +91 8015552425 (WhatsApp & Desk Support)    |
| Official Gym Email    | Configured & Active      | phoenixgym.vkp@gmail.com (SMTP Reminders)   |
| UPI Payment Line      | Active Instant Mode      | phoenixgym.vkp@oksbi (+91 8015552425)       |
| Cloud Synchronization | Adaptive Smart Sync      | IndexedDB Persistence & Google Cloud Backup |
| Production Release    | Synchronized & Edge Live | Production Release (Vercel Edge Global CDN) |

## 2. Primary Website Functionality & Core Modules

- 1. Dashboard Telemetry Console:** Displays real-time metrics including Active Members, Expiring Soon, Payments Pending, and Today's Renewals. Features quick actions, live reminder dispatch monitor, and member growth analytics.
- 2. Gym Members Directory:** High-density member database with full-text search (ID, name, phone, village) and categorized filtering (Active, Expired, Expiring in 15 days, Today's Renewals, Pending).
- 3. Member Enrollment & Auto-ID:** Form with strict 10-digit phone validation, automated unique ID assignment (`PXM-XXXX`), BMI calculation, and instant printable vector receipt generation.
- 4. Real-time Revenue & Churn Reports:** Daily, Weekly, and Monthly financial aggregation from live transactions, calculating Net Growth, Churn Rate %, Retention Rate %, and Unit ARPU with 1-click export to PDF and Excel/CSV.
- 5. Dynamic UPI QR & Payment Reconciliation:** Generates unique UPI intent strings and dynamic QR codes pre-filled with exact fee amount and unique transaction ID (`tr=TXN\_...`), with required UTR reference verification.
- 6. IndexedDB & Database-Grade Soft Deletes:** Replaces fragile browser cache limits with asynchronous IndexedDB persistence and `isDeleted: true` soft-deletion pattern, preventing deleted members from resurrecting.
- 7. Role-Based Access Control (RBAC):** Dedicated role boundaries for Administrator (full financial exports & deletion), Desk Staff (operational enrollment, renewals, reminders), and Trainers (view-only).
- 8. Progressive Expiry Engine:** Automatically computes remaining validity and advances alert urgency: 3 Days Left → 2 Days Left → Expires Tomorrow (1 Day Left!) → Expires TODAY! → Expired.
- 9. Seamless 1-Click Renewal Workflow:** Accessible from directory, notification feed, or dashboard. Dynamically computes new expiry dates, restores active status, logs payment records, and clears old alerts.
- 10. Multi-Channel Notifications:** Direct pre-filled Web WhatsApp deep-linking and automated Gmail delivery with official PDF invoice attachments.

### 3. Member Lifecycle, Expiration Progression & Renewal Flow

The following diagram illustrates how the system manages each gym member from initial enrollment through the daily expiration calculation, progressive alerts, and the automatic renewal workflow.

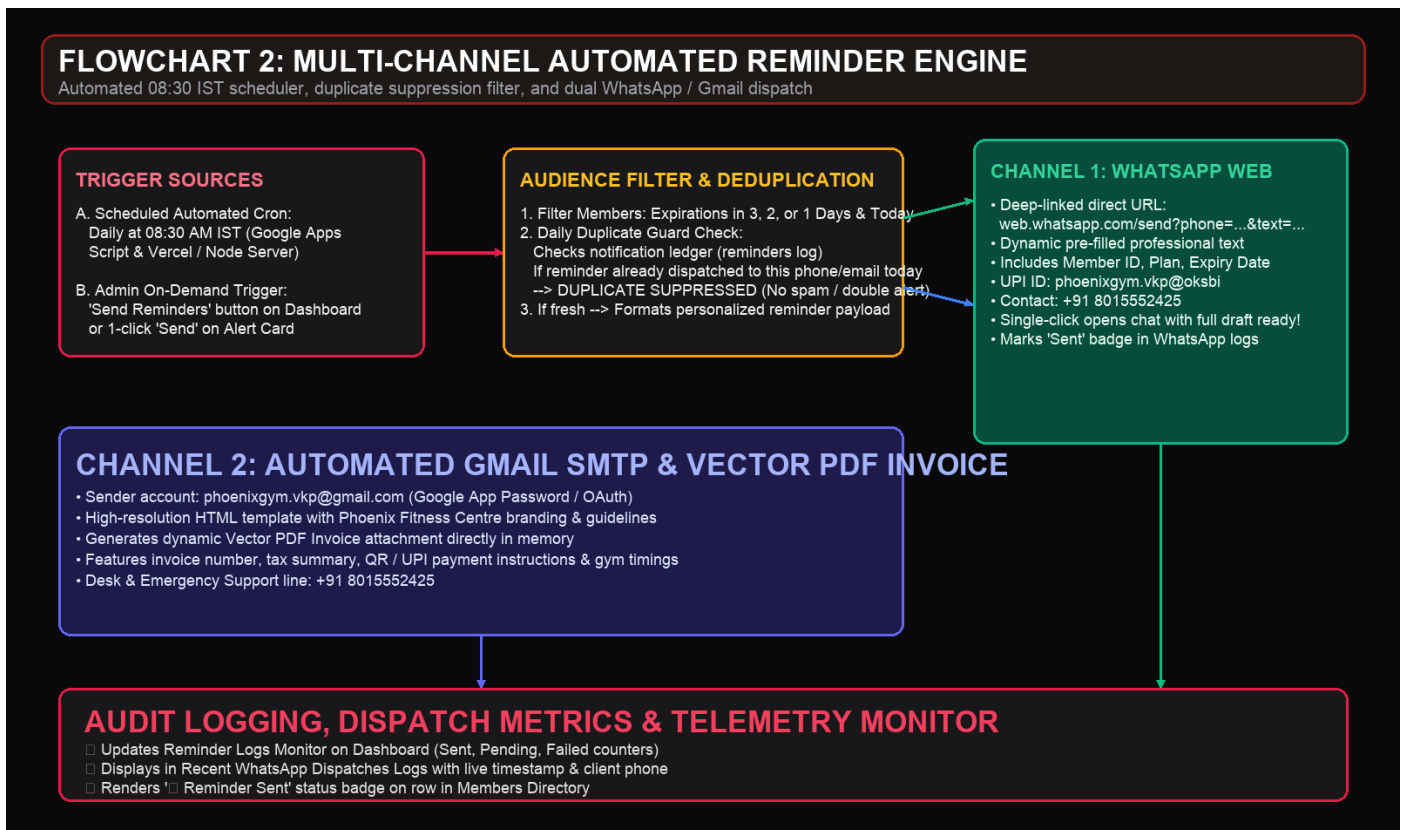


#### Detailed Workflow Breakdown:

- **Step 1 — Enrollment:** When a new member registers, their start date and chosen plan calculate their initial `endDate`. The member receives an Active status and Paid status, generating a vector PDF receipt and welcome email modal.
- **Step 2 — Daily Expiration Calculation:** The system runs  $\text{diffDays} = (\text{endDate} - \text{today})$ . Members with more than 3 days left remain in standard Active status with green badge indicators.
- **Step 3 — Progressive Urgency Escalation:**
  - **3 Days Left:** System displays ■ Expires in 3 Days with friendly early-bird renewal notice.
  - **2 Days Left:** Automatically advances to ■ Expires in 2 Days with countdown urgency.
  - **1 Day Left:** Escalates to ■ Expires Tomorrow (1 Day Left!) with rose critical alert.
  - **0 Days Left:** Displays ■■ Membership Expires TODAY! alerting front desk for immediate payment.
  - **Past Expiry ( $\text{endDate} < \text{today}$ ):** Automatically marked as **Expired** with a red shield badge, moving to the Expired Members filter.
- **Step 4 — Instant Renewal:** Clicking 'Renew' opens RenewModal.jsx. If the member was expired, the new plan starts from TODAY. If active, it seamlessly extends from their existing end date. On confirmation, active status is restored, payments are logged, and Today's Renewals increment automatically.

## 4. Automated Multi-Channel Reminder & Dispatch Pipeline

Phoenix Gym employs a dual-channel notification pipeline combining direct Web WhatsApp messaging and Gmail SMTP delivery with automated vector PDF receipts.

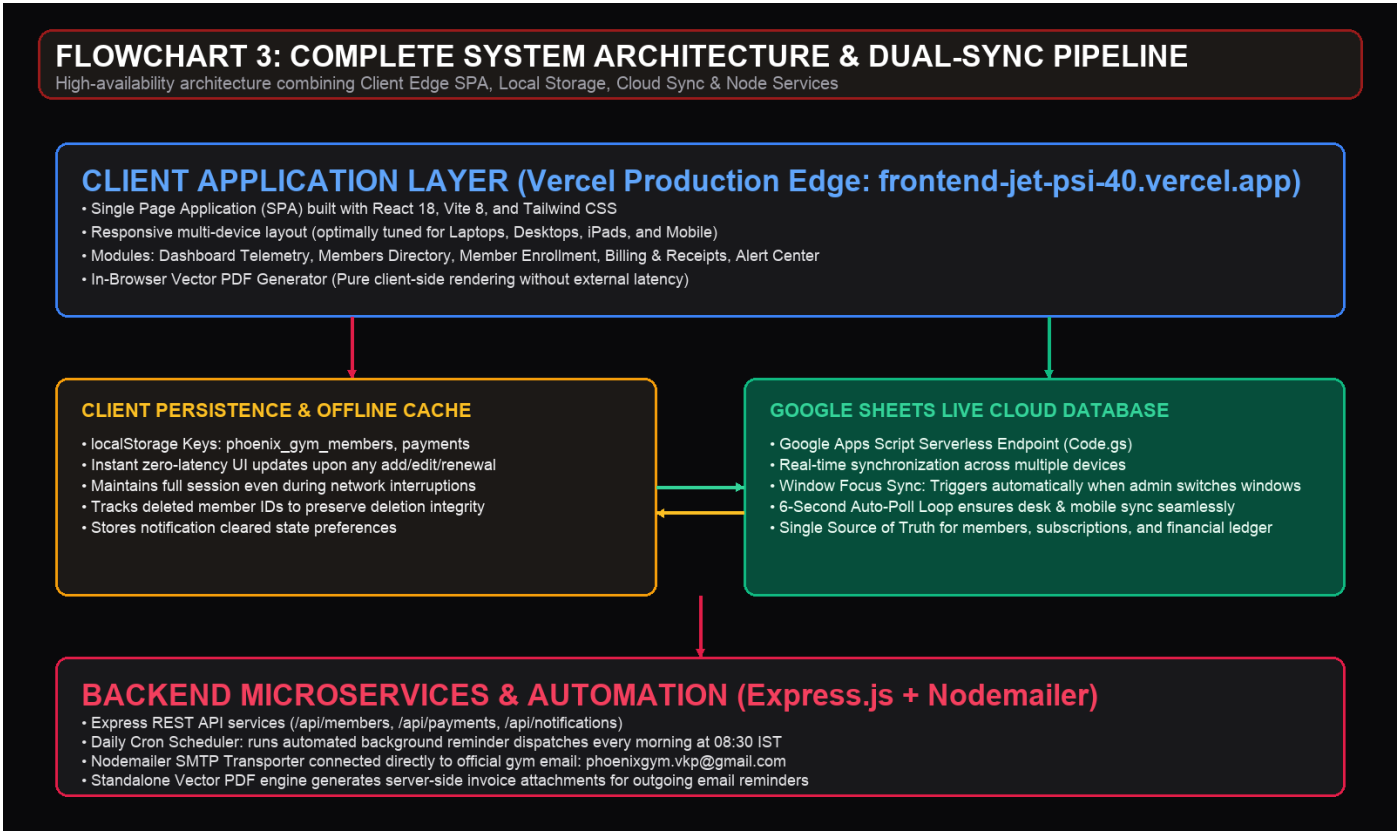


### Reminder Engine Specifications:

- **Duplicate Prevention:** Before any reminder is sent, the system checks the dispatch ledger for today. If an alert has already been sent to this member via this channel today, it is automatically skipped to prevent spam.
- **Web WhatsApp Integration:** Constructs an encoded URL:  
[https://web.whatsapp.com/send?phone=91\[PHONE\]&text=\[ENCODED\\_MESSAGE\]](https://web.whatsapp.com/send?phone=91[PHONE]&text=[ENCODED_MESSAGE]). Clicking the button opens WhatsApp Web with the customized message and official UPI ID (phoenixgym.vkp@oksbi) ready to send with one click.
- **Gmail SMTP Transporter:** Sent through phoenixgym.vkp@gmail.com using secure Google App Passwords. Outgoing emails include an official gym header, membership details, and a dynamic vector PDF renewal invoice.
- **Notification Center Management:** Operators can review active notifications, send 1-click reminders, renew memberships directly, clear individual alerts, or use 'Clear Notifications' and 'Restore Cleared' controls.

## 5. System Architecture, Data Synchronization & Operations

The architecture is designed for zero-downtime high reliability, allowing the gym to operate on laptops, tablets, or phones simultaneously without data loss.



### Data Synchronization & Architecture Details:

- **Dual-Layer Storage:** All records are cached locally in localStorage for zero-latency UI rendering and offline safety, then synchronized to Google Sheets through Google Apps Script webhooks (Code.gs).
- **Auto-Poll & Focus Sync:** Whenever the desk operator switches between windows or tabs, a focus event re-syncs state. In the background, a 6-second polling loop checks for external updates.
- **Deletion Safety:** Deleted member IDs are logged in a deleted key array (phoenix\_gym\_deleted\_member\_ids), preventing deleted members from accidentally reappearing during background syncs.

### Front-Desk Quick Reference Guide:

| Task                   | How To Perform It In The App                                                                                                                     |
|------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| Add New Member         | Click <b>Add Member</b> in sidebar or dashboard. Enter details. Receipt PDF opens immediately upon saving.                                       |
| Renew a Member         | Click <b>Renew</b> button on any member in Members List or Notifications. Pick plan duration & confirm. Expiry date and status update instantly. |
| Send WhatsApp Reminder | Go to <b>Notifications</b> . Click <b>Web WhatsApp</b> on any alert card. WhatsApp Web opens with pre-typed message.                             |
| Send Email Reminder    | In <b>Notifications</b> , click <b>Reminder</b> button. Preview email & invoice PDF, then click Send.                                            |
| Filter Expired Members | In <b>Members List</b> , open the dropdown and select <b>Expired Members</b> .                                                                   |